

Tail Recursion Case Studies

Let's examine some of the recursive functions demonstrated earlier in this book to see if they are good candidates for tail recursion. Keep in mind that because Python and JavaScript do not actually implement tail call optimization, these tail recursive functions will still result in a stack overflow error. These case studies are for demonstration purposes only.

Tail Recursive Reverse String

The first example is the program to reverse a string that we made in Chapter 3. The Python code for this tail recursive function is in *reverseStringTailCall.py*:

Python

```
❶ def rev(theString, accum=''):
    if len(theString) == 0:
        # BASE CASE
        ❷ return accum
    else:
        # RECURSIVE CASE
        head = theString[0]
        tail = theString[1:]
        ❸ return rev(tail, head + accum)

text = 'abcdef'
print('The reverse of ' + text + ' is ' + rev(text))
```

The JavaScript equivalent is in *reverseStringTailCall.html*:

JavaScript

```
<script type="text/javascript">
❶ function rev(theString, accum='') {
    if (theString.length === 0) {
        // BASE CASE
        ❷ return accum;
    } else {
        // RECURSIVE CASE
```